

Regime 46: Volatility Clustering Q2

EURUSD · 60-minute bars (H1) · n=25,000 bars

Symbol: EURUSD

Bar timeframe: 60 minutes (H1)

Total bars in backtest sample: 25,000

Bars with this regime (primary label): 30 (0.12% of sample)

Quadrant: Q2

Win rates (wr_1r-wr_4r): fraction of trades where price reached $\geq 1R \dots \geq 4R$ favorable excursion before ATR-based stop (see scorecard script).

rank_score: blends wr_1r-wr_4r and lightly penalizes strategies with trades < 8 .

Next pages: Regime52Params (classifier + indicator windows), this regime's taxonomy line, then each strategy's name, signal_kind, side rule, and numeric thresholds.

Regime detection — parameter terms and values (Regime52Params)

EURUSD · 60-minute bars (H1) · n=25,000 bars

Regime52Params — shared by the 52-regime classifier and signal indicators:

```
adx_period: 14
rsi_period: 14
ema_slow: 50
ema_ob: 20
ema_fast: 12
structure_lb: 10
mom_lookback: 60
bb_period: 20
z_period: 20
squeeze_win: 250
vol_cluster_win: 48
hmm_vol_quantile: 0.75
```

Classifier rules (thresholds per regime id, claim order, optional ctx_* columns) live in forex_regime/regimes52/classify.py — add context columns to activate macro/event regimes when those rows have data.

This regime (id 46) — taxonomy terms
EURUSD · 60-minute bars (H1) · n=25,000 bars

Section: SECTION 9 — Quant / statistical

Narrative / rule summary:
Volatility Clustering — persistence in |returns|; GARCH family models

Strategies for regime 46 — names, keys, sides, signal values

EURUSD · 60-minute bars (H1) · n=25,000 bars

Sample: EURUSD · 60-minute bars (H1) · n=25,000 bars

Strategy rows for this regime (terms). Entries are counted only when regime52_id matches this regime; signal math is shared across all regimes in the same quadrant— only labels differ.

- R46-S01: Stress mean reversion (tail hedge tilt) — Volatility Clustering
signal_kind='fade_extreme_rsi'; side=-1 → only bearish rule arm fires short
- R46-S02: Wide-stop directional breakout — Volatility Clustering
signal_kind='squeeze_breakout_dir'; side=0 → model picks long or short from rules
- R46-S03: Crash / momentum reversal scalp — Volatility Clustering
signal_kind='impulse_reversal'; side=-1 → only bearish rule arm fires short
- R46-S04: Trend-with-vol overlay (short side bias ready) — Volatility Clustering
signal_kind='breakout_with_trend'; side=-1 → only bearish rule arm fires short

Signal kinds — numeric values and windows (same for every regime using that kind):

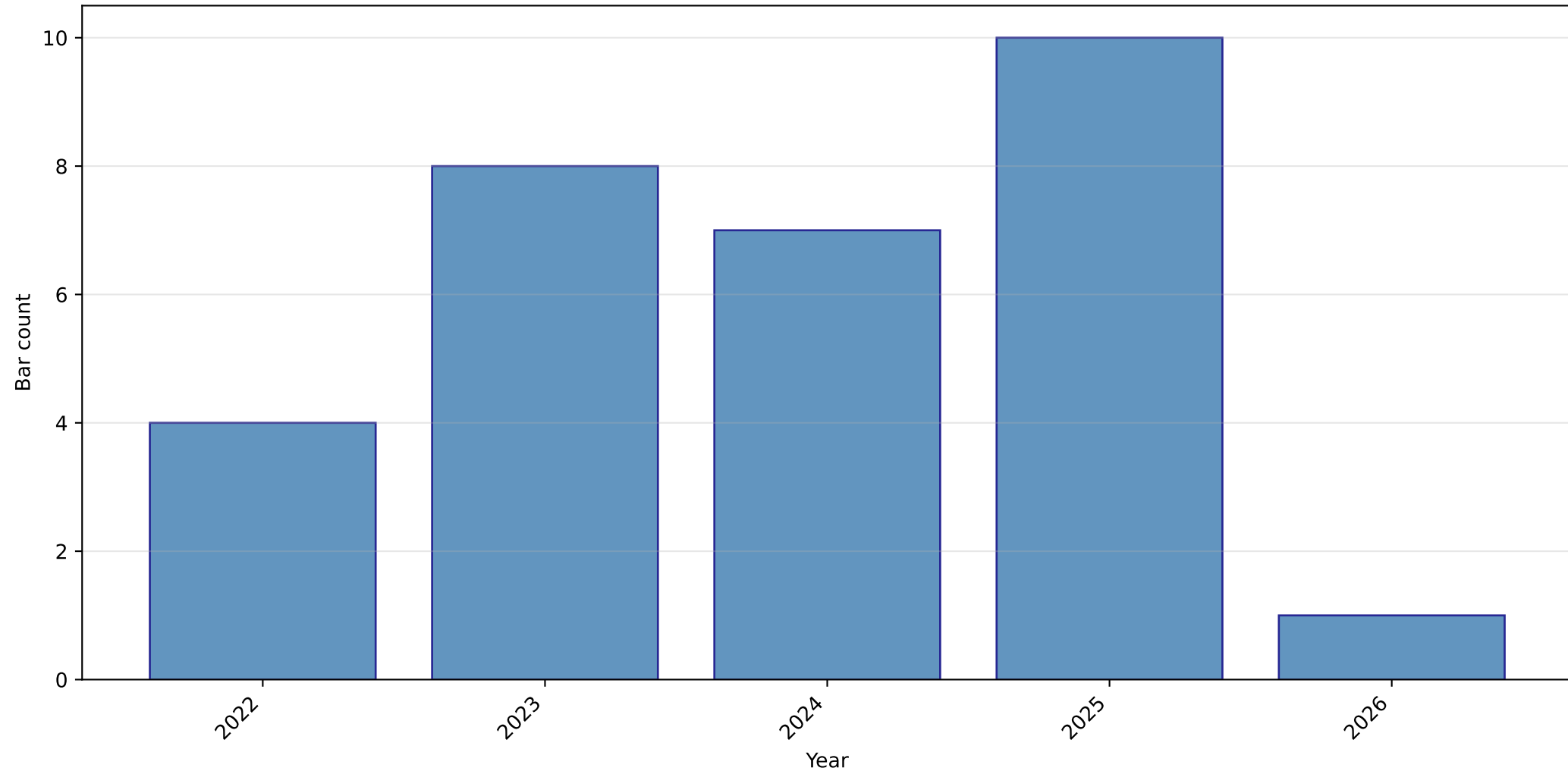
[fade_extreme_rsi]
RSI period=14: long if RSI < 32; short if RSI > 68 (sideRule trims which arm fires)

[squeeze_breakout_dir]
BB width = (upper-lower)/middle on BB from bb_period=20
Tight width: < rolling 100-bar quantile 0.28 (min_periods=40)
Impulse bar: (high-low) > 1.15×ATR (ATR period=14)
Direction: vs ema_slow=50 and candle color (side=0: both dirs)

[impulse_reversal]
3-bar return < -0.008 & RSI < 42 → long; 3-bar return > 0.008 & RSI > 58 → short
(RSI period=14)

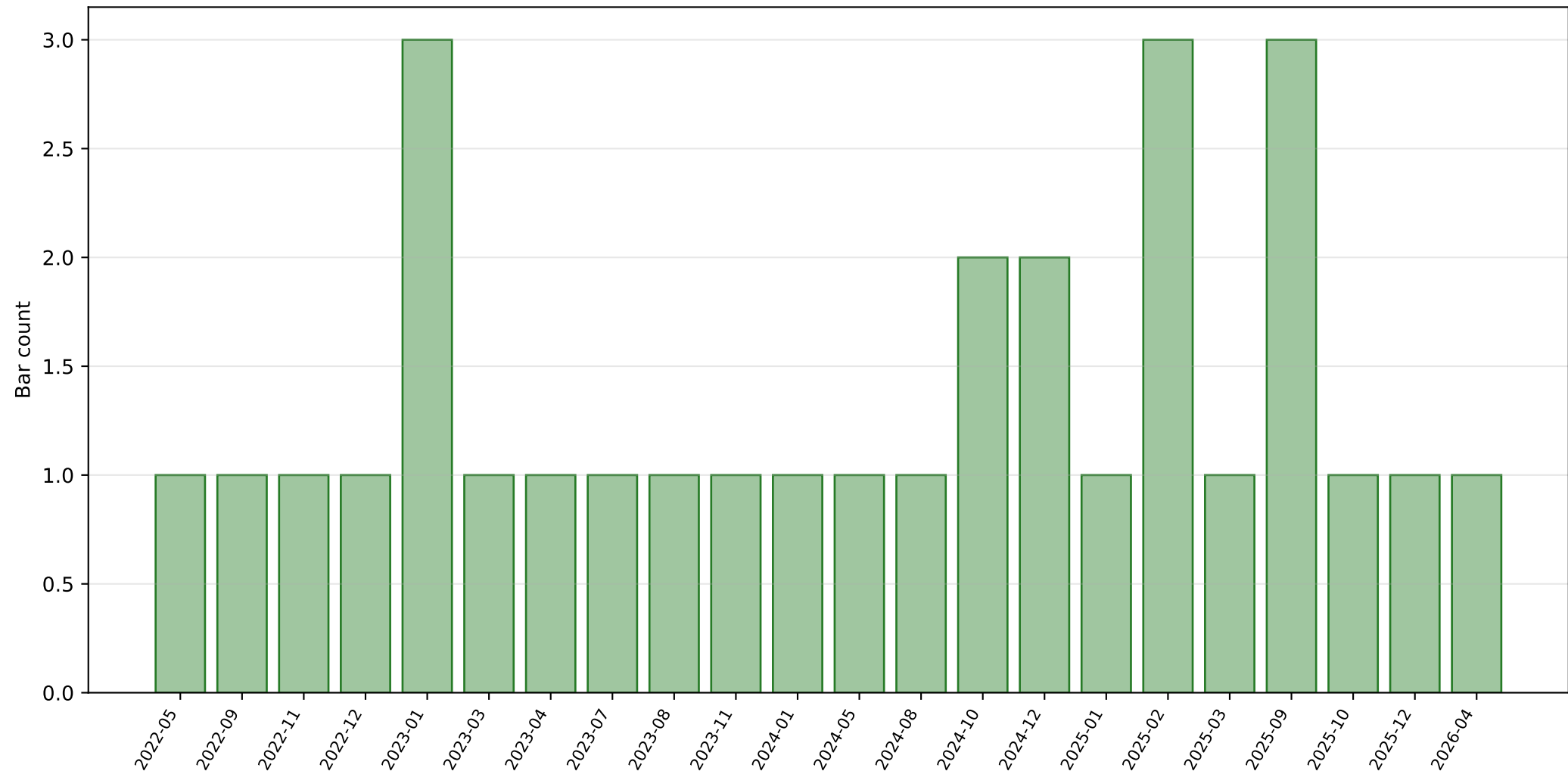
[breakout_with_trend]
Donchian high/low: rolling max(high) / min(low), period=20
EMA filter: close vs ema_slow (period=50)
Long: close > prior Donchian high & close > ema_slow
Short: close < prior Donchian low & close < ema_slow

Regime 46 — bar counts by year (UTC)
EURUSD · 60-minute bars (H1) · n=25,000 bars



Regime 46 — bar counts by month (UTC, last 22 months)

EURUSD · 60-minute bars (H1) · n=25,000 bars



Top strategies by rank_score — regime 46
EURUSD · 60-minute bars (H1) · n=25,000 bars

